Nike Global Test Method: G035, Version 7

Last Version Date: September 7, 2022

Nike Global Test Method G035 –

Standard Test Method for Washability, Feel and Look

1. Summary

1.1 This test determines the effect of cold-water washing on the "FEEL" and/or the "LOOK" of a material or shoes by washing three specimens for 5 cycles in a washing machine with detergent and water, assessing for changes in "LOOK" after each cycle and "FEEL" after the specimens have dried.

1.2 The test is performed on shoe upper materials, flat samples, or whole shoes.

2. Significance and Use

2.1 This test determines the effect of machine washing on a material or shoes.

2.2 This is a qualitative quality assurance test that uses visual and tactile observations to predict potential changes to shoe parts as a result of washing by the customer. Changes to the look and feel of the test specimen are reported in order to highlight areas of possible long-term quality concerns such as color fading, color bleeding, changes to coatings, and bond delamination.

3. Sample Requirements

3.1 For components, see the table below:

Component Type	Component Notes	Minimum Specimen Quantity	Total Specimen Area (metric)/Accepted Shoe Sizes	Minimum Dimensions to meet Requirements (metric)
Flat textile sample, Shoe upper material	-	4 (3 test, 1 control)	871 cm^2	Length: 19.1 cm
Shoe	-	4 (3 test, 1 control)	All shoe sizes accepted	-

**Total specimen area describes the sum of the area of the specimens and does not account for die cutting inefficiencies. Minimum dimensions to meet requirements include 1 cm on the length and width to allow for clean cuts.*

4. Version History

Version #	Date	Author	Reason
1	May 09,1991	-	-
2	August 23, 2005	Gabriela Lungu	Updated formatting, changed the soil release test to a custom test as specified in the Appendix, and changed the initial test method to evaluate the effect of cold water washing on the "FEEL" and "LOOK" of materials or shoes.
3	October 11, 2006	Nancy Felten	Addition of sample size.
4	September 04, 2007	Lam Pham	Correction of outline numbering in section 7.4.1
5	April 26, 2017	Gabriela Lungu	Replace Purex Detergent with Kirkland Signature Detergent.
6	January 28, 2022	Ingrid Hagen-Keith, Jennifer Polivka,	Moved test method content to new format which introduced the following sections:

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		Serene Shehabi, Naiara Hlebczuk	1. Summary (was Overview) 2. Significance and Use 3. Sample Requirements (was Sample Size) 4. Version History (was Revision History) 5. Referenced Documents 6. Terminology 7. Instruments (was Set-Up & Adjustment of Test Equipment) 8. Consumables and Supplies (was Appendix 8) 9. Hazards 10. Preventative Maintenance and Routine Calibration 11. Sample Conditioning 12. Specimen Preparation (was Specimen Preparation) 13. Instrument Preparation (was Set-Up & Adjustment of Test Equipment) 14. Procedure (was Data Collection & Analysis) 15. Calculation and Interpretation of Results 16. Report Some content has been re-ordered into the appropriate section, clarified, or re-worded. Content that has been added and is entirely new includes: - Re-phrasing of summary section and re-ordering of content - Added relevance to footwear in 2.2 - Added total specimen area, shoe size, and minimum dimensions to meet requirements to sample requirements (was sample size). - Content associated with sections: 5. Referenced Documents, 6. Terminology, 7.2 Instrument, 8.2-8.4 Consumables & Supplies, 9. Hazards, 10. Preventative Maintenance and Routine Calibration from T2 forms, 11. Sample Conditioning, 13.1 Instrument Preparation, 14.4-14.6 for calculating and measuring detergent. Content that includes notable clarification include: - Re-phrasing of summary section and re-ordering of content (was overview). - Changed “revision” to “version” to reflect changes in technology. All documents that started with revision 0 now start with 1, and thus all following versions reflect this numbering. - Summary of procedure. - Clarified that multiple samples should not be washed together to prevent cross-contamination in 14.2. - The following content has been removed with the agreement of GLT: 7. Appendix – Custom Soil Release Test Method: this content has been documented as a Beaverton method to reflect its current use.

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			- Changed Analytical Scale to Standard Scale and added specifications based on T2 Supplier Lab Audit May 2021 form
7	September 7, 2022	Cass Sims-Knight	Added ISO reference link for lab conditions, fixed typos.

5. Referenced Documents

5.1 Basis Documents – these documents were referenced during development of this test method

5.1.1 ISO 18454:2018 Footwear – Standard Atmospheres for Conditioning and Testing of Footwear Components for Footwear

5.1.2 T1 Supplier Lab Audit Scorecard (Nike link/external link)

5.1.3 T2 Supplier Lab Audit Form (Nike link/external link)

5.1.4 PPM SOP for T2 Supplier (Nike link/external link)

5.2 Mentioned Documents – these documents will need to be read in order to properly run this test method

5.2.1 None

5.3 Best Practice Documents – this test method is referenced in the following best practices

5.3.1 Fuse (Nike link/external link)

5.3.2 Flyknit (Nike link/external link)

5.3.3 Sockliner (Nike link/external link)

6. Terminology

6.1 *Control:* A specimen where the factor being tested is not applied that is used as a point of comparison to the tested specimens. For example, in washability, the factor is the effect of washing the sample in a washing machine. The control is a specimen that has not been tested in the washing machine (or any other test).

6.2 *Feel:* The way a test specimen feels to the touch. This is a qualitative measurement that includes, but is not limited to, softness, stickiness, and suppleness.

6.3 *Look:* The way a test specimen appears visually. This is a qualitative measurement that includes, but is not limited to, cracking finishes, bond delamination, shrinking, color bleeding, fraying, and broken threads.

6.4 *Minimum specimen size:* A minimum number of specimens required to represent the population accurately for a given test. For example, a test requires two shoes from a pair of Air Force Ones to represent all Air Force Ones produced on a given day at a given factory.

6.5 *Sample:* A group of specimens that are taken from a larger population for measurement. For example, 2000 Air Force Ones pairs are produced on a given day at a given factory – this is the population. One pair of Air Force Ones from this population is a sample.

6.6 *Specimen:* A specific item for testing. For example, within the received Air Force One sample, there are 2 shoes. Each individual shoe is a specimen.

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7. Instruments

7.1 Programmable Washing Machine. The preferred brand is Whirlpool, but other brands are acceptable.

7.1.1 Washing machine must have a cool wash setting (27°C ± 3 °C).

7.1.2 Each wash-rinse-spin cycle should take approximately 25 minutes.

7.2 Standard Scale to precision of 0.001 grams

8. Consumables and Supplies

8.1 Kirkland Signature Heavy-Duty Laundry Detergent – 28 lb bucket

8.1.1 Laundry Detergent supplied by OIA



Figure 8-1: The detergent required for testing.

8.2 Ruler with precision to 1 mm

8.3 Scissors

8.4 Permanent marker that is a different color than the specimen

9. Hazards

	Warning	Instrument	Risk of Loss	Mitigation
MACHINE	 GENERAL WARNING SLIP HAZARD	Washing Machine	Risk of skin irritation, back injuries, burns, and noise Risk of slips, trips, and falls due to water on the ground	Use gloves for skin protection. Avoid awkward reaching. Squat, don't bend. Be aware of potential hazards and keep alert to protect your safety. Be careful on wet surfaces. Clean up spills.
CHEMICAL	 WASH HANDS	Kirkland Signature Heavy-Duty Laundry Detergent	Risk of serious irritation if it comes in contact with the eyes	Wash hands thoroughly after handling. If product gets in eyes, rinse thoroughly with water for at least 15 minutes.
PROCEDURAL	N/A			

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10. Preventative Maintenance and Routine Calibration

10.1 Annually, ensure the calibration of the Standard Scale is being done.

10.2 Weekly, ensure that the leveling and cleaning of the Standard Scale are being done.

10.2.1 Check the level indicator on the scale is centered. If the level indicator is not centered, adjust the footpad/rest.

10.2.2 Use dry soft cloths to dust the scale. Use a cloth moistened with water to gently clean dirty fingerprints and smudges (do not use solvents or harsh detergents for cleaning).

11. Sample Conditioning

11.1 Samples must sit for 24 hours after assembly before sample conditioning.

11.2 Samples must sit at 23±2°C for at least two hours prior to specimen preparation.

12. Specimen Preparation

12.1 *Flat Textile Sample:* If a large swatch of material is provided, using a ruler, marker, and scissors, measure and cut four 11.4-cm-by-19.1-cm test specimens from it.

12.2 *Uppers:* No additional specimen preparation is required.

12.3 *Shoes:* No additional specimen preparation is required.

13. Instrument Preparation

13.1 Inspect the interior of the washing machine to ensure the drum is empty and clean.

13.2 Set the water temperature to the **Cool** setting (water temperature is 27±3°C).

13.3 Set the water level to **Small Load** or **Low Level**.

13.4 Select the **Regular Fabrics** wash cycle.

13.4.1 This wash cycle must involve normal wash, rinse, and spin cycle that takes approximately 25 minutes to complete.

13.4.2 For WHQ washing machine: Set machine to division 12 on regular fabrics.

14. Procedure

Summary: Place up to 3 shoes (1.5 pairs of shoes), flat textiles specimens (11.4 cm by 19.1 cm), or shoe uppers inside the washing machine. Wash the specimens with detergent five times in a regular, small load at the cool setting. Evaluate and document any changes to look after each cycle. After the 5th cycle, lay specimens on a flat surface to air dry. Evaluate and document changes in feel once specimens have dried.

14.1 Set aside one specimen. This will be the control specimen that will be compared to the washed specimens.

14.2 Place test specimens for a single sample into the washing machine. Multiple specimens from the same sample can be washed together, but multiple samples cannot. Each sample must be tested separately to prevent cross-contamination during testing.

14.2.1 Flat samples or shoe uppers: Do not fill the machine more than 33% full.

14.2.2 Whole shoes: A maximum of 3 shoes (1.5 pairs) can be tested per wash cycle.

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14.3 Calculate the amount of detergent needed such that there is 1.5 grams of Kirkland Signature Heavy-Duty Laundry Detergent per liter of water in the wash cycle (1.5 g/L) rounded to the nearest gram with no decimals using the following formula:

$$Detergent\ (g) per\ wash = 1.5g\ detergent \times Liters\ of\ water\ per\ small\ load$$

14.3.1 Example: Nike WHQ's Washing machine uses 33 L of water per Small Load: 1.5 g/L x 33 L = 50 g detergent per wash (rounded to nearest whole number) which is equivalent to ½ cup of detergent.

14.4 Place the container for the detergent on the standard scale and zero the balance.

14.5 Weigh the calculated required detergent into the container.

14.6 Add the detergent to the washing machine.

14.7 Start the washer and allow the full wash cycle to complete.

14.8 Once the wash cycle has finished, remove the damp specimens and examine for any changes in look (cracking or flaking of the finish, bond delamination, shrinkage, color bleeding, fraying, broken threads, etc.) by comparing it with the control specimen. Record the cycle number and any changes that are seen.

14.9 Repeat Steps 14.2–14.5 until a total of 5 wash cycles have been completed.

14.9.1 If a specimen shows progressive changes in appearance during testing, make notes or take pictures to qualify the change.

14.10 At the end of 5 cycles, remove specimens from the washing machine.

14.10.1 Flat textile samples or shoe uppers: Lay face up on a flat surface or hang specimens and allow to air dry.

14.10.2 Whole shoes: Place the outsole of the shoes on a flat surface and allow to air dry.

14.11 After the specimens have dried, examine for any changes in look (including but not limited to: cracking or flaking of the finish, bond delamination, shrinkage, color bleeding, fraying, broken threads, etc.) and feel (including but not limited to: softness, stickiness, and suppleness) by comparing it with the control specimen. Document the change with pictures.

15. Calculation and Interpretation of Results

15.1 Not applicable.

16. Report

16.1 Report and describe any change in "LOOK" after each wash cycle like finish coating peeling, bond delaminating, shrinkage, color bleeding, fraying, and/or thread broken. Pictures are encouraged.

16.2 Report and describe any change in "FEEL" following air drying after wash cycle 5 like changes in softness, stickiness, or suppleness. Pictures are encouraged.

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